

CST: A Decentralized Real-World Asset Standard

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Abstract. A tokenized stock should not require you to trust a single custodian, nor to lock three dollars of collateral to hold one dollar of exposure.

Two models dominate onchain real-world assets (RWAs), and each fails a different half of the problem. **Custodial tokens** (Ondo, Backed/xStocks, Robinhood) track the asset faithfully but concentrate the entire claim on one intermediary: an issuer, a broker-dealer, a prospectus you must trust. **Collateralized-debt-position (CDP) synthetics**, the model that made stablecoins like DAI work, are trustless, but break for RWAs. A stablecoin CDP carries only *one* volatile leg: the collateral moves, but the liability is pinned at \$1, so the system only has to buffer swings on one side. An RWA synthetic carries *two*. The minted token tracks a moving market price while the collateral behind it moves independently. Backing one volatile asset with another, imperfectly correlated, forces punitive overcollateralization to stay solvent, and that capital does not scale.

We introduce the **Collateral-Secured Token (CST)**: an ERC-20 that tracks a real-world asset and is backed by a *diversified portfolio of collateral* under a single code-enforced solvency ledger, the way MakerDAO backs DAI with a diversified collateral set rather than a bank account. The first layer is a protocol-owned **Reserve Vault** per asset, holding existing tokenized stocks 1:1 by value and convertible in both directions through a **peg-stability module (PSM)**; a fully reserved token is delta-one backed and consumes no other capital. The second layer is a pool of onchain **Multi-Purpose Vaults (MPVs)**: overcollateralized crypto collateral posted by liquidity providers that insures any exposure not matched by reserves, and simultaneously underwrites a lending market and earns native yield. Market makers quote mints and redeems through an RFQ marketplace, and every eToken carries a code-property exit: a maker fill, an in-kind PSM redemption against the reserve, or a permissionless forced redemption against vault collateral at the oracle price. In future phases the same ledger admits further backing sleeves, including attested offchain hedges, decentralizing the portfolio issuer by issuer. This paper specifies the CST, the backing stack, the RFQ and PSM issuance paths, the pooled solvency accounting, the redemption guarantee, and the trust model, semi-technically, with the mechanism design made explicit.

1. Introduction

Commerce on the internet has come to rely on trusted third parties [1] to hold the real-world assets that onchain tokens represent. The tokenization of stocks, bonds, and commodities works well enough for

most transactions, but it still suffers from the inherent weakness of the custodial model. A token that says *eTSLA* is only as good as the entity that promises a share of Tesla sits behind it: its broker-dealer, its jurisdiction, its prospectus, its solvency.

What is needed is a way to hold real-world-asset exposure onchain that does not depend on trust in any single custodian, and that a holder can verify directly. Crypto already solved this problem once, for the dollar. DAI [3] did not eliminate custodial assets; it *diversified* them, backing one token with a portfolio of collateral (crypto, custodial stablecoins through a peg-stability module, eventually real-world credit) unified by a solvency rule enforced in code. No single collateral type, and no single issuer, is load-bearing. The instrument that got DAI started, the **collateralized debt position (CDP)**, is however the wrong tool to reuse for stocks, and understanding *why* is the foundation of this protocol.

Two families of solution exist today, and they fail on opposite axes:

MODEL A · CUSTODIAL

Backed tokens

Ondo, Backed / xStocks, Robinhood. A regulated issuer holds the physical share and issues a claim. **Tracks perfectly, but you must trust the one custodian:** counterparty risk, jurisdiction, opacity, redemption at the issuer's discretion. Useful as an ingredient; fragile as the whole answer.

MODEL B · CDP SYNTHETIC

Overcollateralized synthetics

The DAI model applied to an RWA. **Trustless, but breaks for stocks:** it backs a volatile asset with another volatile asset, so the collateral it demands scales with the very volatility you want to hold. Great for a \$1 peg; wrong for a moving price.

Own Protocol proposes a third model that does to the stock token what DAI did to the dollar: keep the tracking fidelity of the custodial issuer, but demote every custodian to one substitutable ingredient in a diversified, code-governed collateral portfolio. We call the instrument a **Collateral-Secured Token**. The remainder of this paper builds it up from first principles: first the failure of the CDP for RWAs (§2), then the CST and the backing stack behind it (§3), the two vault types that hold that backing and the lending market they underwrite (§4), the two issuance paths (§5), how the system stays solvent (§6–7), how the incentives compound (§8–9), who the system serves (§10), the oracle and the trust model (§11–12), and the path by which the portfolio decentralizes (§13).

2. Why CDPs fail for RWAs

A CDP mints a synthetic token against locked collateral and relies on liquidation to keep it solvent. For a stablecoin this works beautifully. For a real-world asset it breaks, and the deeper reason is not tracking but capital: a CDP backs a volatile asset with another volatile asset, and the cost of doing that safely grows with the very volatility people want to hold.

2.1 One volatile leg, or two

A CDP's collateral ratio has to cover the worst-case adverse move over a liquidation window, so what decides its capital cost is simple: how many things in the system can move at once. To mint debt value D the user locks collateral value C at a minimum ratio ρ :

$$C \geq \rho D \quad \implies \quad e = \frac{D}{C} \leq \frac{1}{\rho}.$$

Eq. 1: capital efficiency of a CDP

A stablecoin has *one* volatile leg: the collateral (say ETH) moves, but the liability is pinned at \$1. An RWA synthetic has *two*: the liability tracks a moving market price while the collateral moves independently, and without a hedge nothing nets one leg against the other. What that costs is not a matter of taste. ρ is set by the worst joint move a position must survive *through liquidation*: over that horizon let δ_c be the worst credible collateral drawdown and δ_e the worst credible rise in the liability. Solvency through liquidation requires $C(1 - \delta_c) \geq D(1 + \delta_e)$, i.e.

$$\rho \geq \frac{1 + \delta_e}{1 - \delta_c}.$$

Eq. 2: the collateral ratio is set by the joint worst case

One leg. For a stablecoin the liability is pinned, so $\delta_e = 0$. The binding horizon is not a single liquidation window but a stress episode in which liquidation itself is impaired (congestion, cascading sales, zero-bid auctions: Black Thursday 2020), and ETH has halved in such episodes more than once. $\delta_c = 50\%$ gives $\rho \geq 2$. MakerDAO's parameter floor is 150% [3]; the ratio that survives in practice is $\approx 2\times$, and $e \leq 0.5$.

Two legs. For an RWA the liability moves too, and with no hedge the position must provision for the joint worst case. Keep $\delta_c = 50\%$ and take $\delta_e = 30\%$ (single names gap on earnings and halts, and hot names rally that much in weeks): $\rho \geq 1.3/0.5 = 2.6$ before the liquidation bonus and slippage. Call it $3\times$, and $e \leq 0.33$. And it scales the wrong way: δ_e grows with exactly the volatility people want to hold. In plain terms, **a CDP asks the person who wants \$1 of Tesla exposure to lock up \$3, and the tax is heaviest for the very names that drive demand.** That is negative leverage on a product whose entire appeal is exposure.

Capital is the structural failure; tracking is the practical one. Nothing in a CDP connects the synthetic's onchain price $\tilde{P}_t$ to the true price P_t except speculative flow, so the two drift. §3 supplies both the missing capital model (a layered portfolio instead of per-position margin) and the missing mechanism (a delta-one reserve with two-way convertibility), and Eq. 2 itself returns in §6.1, inverted, as the sizing rule for the crypto collateral layer: the same bound over a bounded claim window instead of an open-ended horizon.

2.2 Reflexivity under stress

Finally, CDP liquidations are reflexive. When the collateral and the minted asset are correlated, as they are whenever crypto sells off, a price drop triggers liquidations, whose forced selling drives the price down further, triggering more liquidations. Stablecoins partly escape this because the debt (a dollar) is *uncorrelated* with the collateral. An RWA synthetic enjoys no such insulation. CDPs, in short, gave crypto trustless dollars; they cannot give it trustless stocks. The next section fixes each failure in turn.

Table 1: CDP vs. CST on the three failure axes.

Axis	CDP synthetic RWA	Collateral-Secured Token
Price tracking	Fragile, oracle-dependent	Delta-one reserve + two-way PSM convertibility (§3–5)
Capital efficiency (minter)	$e \leq 1/\rho \approx 0.33$	$\approx 1\times$: minter pays notional; reserves back it 1:1, LPs insure the residual
Who posts safety collateral	The minter (wants exposure)	The reserve (1:1) and the LP (wants yield), roles separated
Stress behavior	Reflexive liquidation spiral	Self-healing liquidation (§8.3)

3. Collateral-Secured Tokens

A **Collateral-Secured Token (CST)** is a tokenized real-world asset backed by a diversified portfolio of collateral under one code-enforced solvency ledger. The design move is a change in what "backing" means: *stop backing every token with one custodian's promise, and stop margining every position on its own; back the whole supply with a layered portfolio that code can verify and liquidate.*

A CDP asks one pool of locked collateral to do everything: provide the exposure, track the price, and guarantee solvency. That is why it charges ρ dollars per dollar and liquidates on every adverse move (§2). A custodial wrapper asks one company to do everything, which is cheaper but concentrates the entire claim on a single balance sheet. A CST layers the jobs across collateral types, each doing the one thing it is best at:

- **The reserve layer provides tracking.** For each asset, a protocol-owned **Reserve Vault** holds existing tokenized stocks (delta-one wrapper tokens from regulated issuers: an Ondo or Robinhood tokenized TSLA behind eTSLA) matched 1:1 *by value* against issued supply, with two-way convertibility through a peg-stability module (§5.2). A fully reserved eToken is delta-one backed: the reserve moves dollar-for-dollar with the liability, so the pair carries zero net market risk. This is the tracking mechanism a CDP structurally lacks (§2.1), supplied onchain and verifiable block by block.
- **The crypto layer provides insurance.** Independently, LPs deposit crypto collateral (stablecoins, ETH, staked ETH) into onchain **Multi-Purpose Vaults**. This pool overcollateralizes whatever slice of exposure the reserves do not match: new supply a maker has not yet backfilled, a wrapper that depegs, an issuer that fails. It is at risk but not spent, so the same capital simultaneously underwrites the lending market and earns yield (§4), and it is liquidated in exactly one scenario: a redemption goes unfilled and a holder forces it at the oracle price (§7).
- **Future layers plug into the same ledger.** The solvency accounting (§6) prices every backing sleeve in USD and nets it against exposure, so new sleeves — additional wrapper issuers, attested offchain hedges run by market makers — can be added without changing the token or the guarantee (§13).

Three consequences follow, and each one is a failure of §1 inverted:

1. **No single custodian is load-bearing.** A wrapper issuer failing is a marked-down reserve, not a broken token: the residual reloads onto the overcollateralized crypto layer automatically (§6), and the

redemption guarantee never lapses. Diversification across issuers and collateral types replaces trust in any one of them: the DAI playbook, applied to stocks.

2. **Holders carry no liquidation risk, and minters pay 1:1.** There is no position to liquidate, only a portfolio to keep solvent. A price move, however violent, liquidates no holder: collateral moves only on redemption or default (Eq. 4). The punitive ρ of Eq. 1 leaves the minter's side entirely; the reserve backs at 1:1 and the safety margin on the residual is posted by yield-seeking LPs (Eq. 3).
3. **The backing stays productive.** CDP backing sits idle against its own position. Here the reserve layer is a working inventory with two-way flow, and the insurance layer underwrites a lending market and earns native yield while it stands guard (§4).

*One custodian is a promise. A portfolio governed by code is a guarantee.
A backed token asks you to trust the single custodian holding the real share. A Collateral-Secured Token spreads the backing across a diversified, onchain-verifiable portfolio — reserves of existing stock tokens, overcollateralized crypto — and gives every holder a permissionless exit against it.*

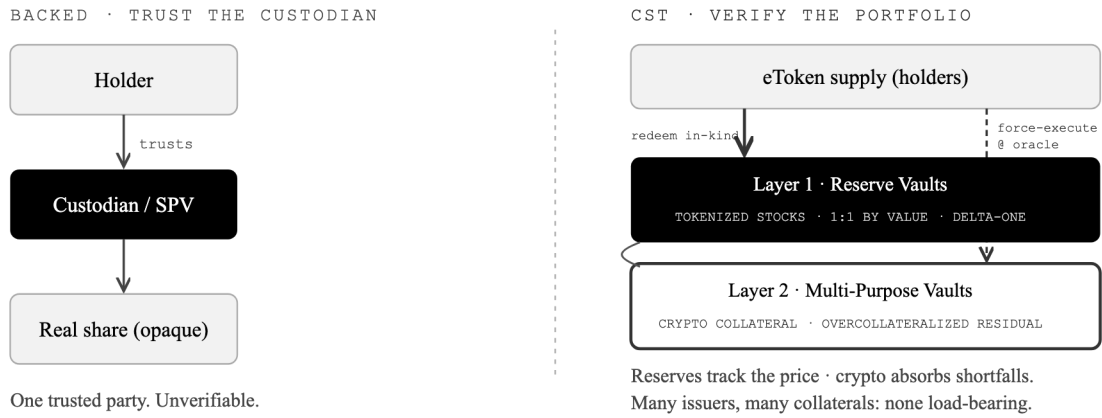


Figure 1. A backed token interposes one custodian the holder must trust (left). A CST is backed by a layered, onchain-verifiable portfolio (right): reserves of existing stock tokens track the price 1:1, overcollateralized crypto vaults absorb any shortfall, and every holder can exit against the stack — in-kind through the PSM or at the oracle price by force execution.

3.1 The backing stack, quantified

Write k for the capital a party must commit per dollar of exposure. A CDP concentrates everything on the minter: $k_{\text{minter}} = \rho \approx 3$, idle by construction, because that collateral is the sole and continuous backing of its own position. A CST splits the ledger across the stack:

$$k_{\text{minter}} = 1, \quad k_{\text{reserve}} = 1 \text{ (reserved slice)}, \quad k_{\text{LP}} = \frac{1}{U_{\text{max}}} \approx 1.5 \text{ (residual only)}.$$

Eq. 3: capital per dollar of exposure, by layer

The reserve's $1\times$ is genuine 1:1 backing, but unlike a custodial wrapper it is held onchain, in a protocol-owned vault anyone can audit, redeemable in-kind by any holder, and substitutable: it is an inventory position, not a bilateral promise. The LP's $1/U_{\text{max}}$ applies only to the slice reserves do not match, and it is not the CDP's ρ wearing a different name; it differs in three ways that compound. It is *pooled*: one

insurance base stands behind every asset and every position, instead of each position dragging its own worst-case buffer. It is *productive*: the pool earns native yield and a lending premium while it stands guard (§9), where CDP backing earns nothing. And it is *voluntary*: posted by parties who are paid a premium to underwrite, not extracted from the person who came for exposure. In steady state, where makers keep assets fully reserved (§5), the marginal dollar of eToken exposure consumes roughly one dollar of productive reserve and only a small crypto buffer — against a CDP's three idle dollars.

The second difference is the trigger. A CDP liquidates whenever the price crosses a threshold. A CST liquidates crypto collateral in exactly one case: a redemption goes unfilled past the claim window.

$$\underbrace{1\{C_t < \rho_{\text{liq}} D_t\}}_{\text{CDP: a price event}} \quad \text{vs.} \quad \underbrace{1\{\text{maker default} \wedge \text{unfilled redemption}\}}_{\text{CST: a credit event}}$$

Eq. 4: what triggers a collateral liquidation

Price events are frequent, correlated with market stress, and reflexive (§2.2): the liquidation itself moves the price that triggers the next one. Credit events are rare and idiosyncratic, the exposure a default can strand is capped at all times by the solvency invariant (§6), and settling a claim shrinks exposure and collateral together instead of dumping inventory on a falling market; §7 proves the resulting stability property (Eq. 13).

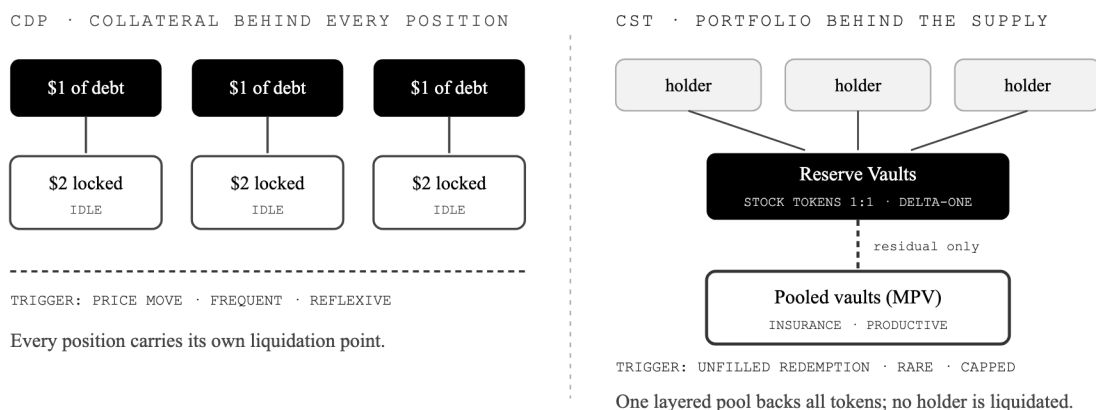


Figure 2. A CDP pins idle collateral behind every position and liquidates on price moves (left). A CST backs the whole supply with a layered portfolio: delta-one reserves first, pooled productive crypto collateral on the residual, liquidated only when a redemption goes unfilled (right). The trigger moves from a price event to a credit event, and holders are never liquidated.

3.2 Neighbors, and why they are not this

The design space around tokenized exposure is crowded, and a CST should be judged against its nearest neighbors, not only against the two failures of §1:

Table 2: adjacent designs, and the axis on which each one breaks.

Design	Trust root	What makes it track	Capital per \$1	Where it breaks
Custodial wrapper (Ondo, xStocks)	Custodian	Real share held 1:1	$\sim 1\times$	One load-bearing intermediary: counterparty risk, opacity, gated redemption. (An ingredient of a CST, not a competitor to it.)
CDP synthetic (DAI model on RWAs)	Code	Nothing structural (§2.1)	$\rho \approx 3\times$, idle	Two volatile legs force punitive, unscalable collateral (§2).
Pooled-debt synths (Synthetix [5])	Code	Oracle exchange against a shared debt pool	$4\times+$ historical	Stakers are the unhedged counterparty to every synth.
Perp DEX (Hyperliquid, Ostium)	Code + venue	Funding-rate convergence	Margin	A position, not a token: funding bleed, liquidation risk, no composability.
Delta-neutral dollar (Ethena [6])	Code + venues	The collateral itself is the hedge	$\sim 1\times$	Hedge and backing share one balance sheet; built for one asset, the dollar.
CST (this paper)	Code	Delta-one reserves + PSM convertibility + oracle-priced forced redemption	$1\times$ minter; $1\times$ productive reserve; pooled LP insurance on the residual	Capacity bounded by collateral; trust model of §12.

The closest architectural neighbor is MakerDAO itself [3]: one liability, a diversified collateral portfolio, a PSM for 1:1 convertibility against custodial assets, and a solvency rule in code. A CST generalizes that architecture from a \$1 peg to a moving price, which is exactly what forces the reserve layer to be delta-one (only an asset that moves *with* the liability can back it 1:1) and the crypto layer to be sized as a claim-window buffer rather than a per-position margin (§6.1). Ethena [6] is the nearest single-asset comparison: a delta-neutral hedge behind a synthetic dollar. But in Ethena the deposited collateral *is* the hedge — one balance sheet, one venue class, one asset — so a hedge failure is a backing failure. In a CST the layers are separable: reserves can depeg, an issuer can fail, and the overcollateralized crypto layer picks up the residual with no change to the token or the guarantee. And where Ethena's insurance fund is protocol-owned surplus, MPV insurance is third-party LP capital that prices itself: underwriting pays a premium, so coverage scales with demand rather than with retained earnings (§9). Synthetix [5] proved pooled collateral can back synths, but asked stakers to be the unhedged counterparty at 400%+ collateralization; a CST nets the price risk against delta-one reserves and leaves LPs only the residual and default risk.

The category has no clean name today (every incumbent is custodial), so we define one. A CST is a mechanism category, like *stablecoin* or *rollup*, not a token interface: at the token layer it is deliberately plain, and everything that makes it a CST lives in how it is issued and secured.

DEFINITION

A token is a **Collateral-Secured Token** if and only if all five hold:

1. **Exogenous target.** It tracks the price of an asset that trades outside the protocol.
2. **Portfolio backing.** Solvency rests on a diversified set of collateral — reserve assets, crypto collateral, further sleeves — under one ledger; no single custodian, issuer, venue, or collateral type is load-bearing, and a failed sleeve reloads onto the others.
3. **Holders are never liquidated.** Backing collateral moves only on redemption or default (a credit event), never on a price move against the holder.
4. **Verifiable solvency.** That the portfolio covers exposure is checkable by anyone, onchain, at any time (Eq. 9).
5. **Permissionless exit.** Any holder can redeem against the backing without anyone's cooperation: in-kind through the PSM, or at the oracle price by force execution (§7).

Fail any one criterion and the instrument is something else; the failures map exactly onto Table 2. A custodial wrapper fails 2, 4, and 5. A CDP synthetic fails 3. A delta-neutral dollar fails 2 (one balance sheet, one venue class). A perp position fails the token premise itself. Concretely, a CST is a standard ERC-20 with ERC-2612 permit [7], one active token per asset, freely composable across DeFi [2]. Holders *mint a CST*, it is *backed by a reserve and secured by an MPV*, and an asset *goes live as a CST*: the vocabulary of the primitive.

4. The Backing Stack: Reserve Vaults & Multi-Purpose Vaults

Two vault types hold the portfolio of §3, and they are deliberately different species. The **Reserve Vault** is protocol-owned backing: stock tokens matched 1:1 against issued supply, with no LP claims on it. The **Multi-Purpose Vault (MPV)** is LP-owned insurance: one pooled collateral base that secures the residual, underwrites the lending market that makes CSTs worth holding, and earns yield while doing both.

4.1 Reserve Vaults: protocol-owned backing

Each asset can register one Reserve Vault per wrapper token: a custody pool holding an existing tokenized stock (an Ondo or Robinhood tokenized TSLA behind eTSLA) as direct backing for that asset alone. Three properties define it, each the opposite of the MPV:

- **It is share-less and protocol-owned.** No LP deposits, no yield queues, no claims. Wrapper tokens enter through PSM mints and permissionless maker deposits, and leave only through PSM redemptions or a surplus skim; every exit is clamped so the remaining reserve still covers the asset's exposure. Redeeming against the reserve transfers no one's wealth: the backing and the liability shrink together.
- **It is delta-one by construction.** The wrapper tracks the same equity the eToken tracks, so reserve value moves dollar-for-dollar with the liability. In the solvency ledger (§6) the reserve nets directly against the asset's exposure: a fully reserved eToken consumes zero crypto collateral.

- **Its surplus pays holders.** Total-return wrappers reinvest dividends, so a matched reserve drifts above the exposure it backs. That surplus is skimmed and distributed to eToken holders through the token's dividend accumulator, passing the underlying asset's income through to the people holding its exposure.

The failure mode is priced in rather than wished away: if the wrapper depegs or its issuer fails, the reserve marks down, the unmatched residual reappears in the global exposure total, and the MPV layer overcollateralizes it automatically (§6). Diversifying reserves across issuers keeps any one such event small.

4.2 Multi-Purpose Vaults: the insurance & lending layer

Start with why lending is not optional. Crypto capital does not park. A holder who cannot redeploy against a position will sell it to fund the next trade, so an RWA with no borrow market loses exactly the users it was built for. Every asset that matters in DeFi (ETH, staked ETH, BTC, stablecoins) matters because a lending market exists for it. For CSTs to become an asset class rather than a curiosity, each one needs a borrow market from day one.

A lending market normally requires its own capital. The MPV's insight is that the insurance pool of §3 is already sitting there: at risk, but not spent. Because its jobs are layered in risk rather than in cash, the same deposited dollar works several non-conflicting jobs at once:

1. **It secures issuance.** The pooled collateral overcollateralizes every dollar of exposure the reserves do not match, and is the fund that pays forced redemptions (§7).
2. **It underwrites lending.** The same collateral underwrites a lending market where CST holders borrow stablecoins against their tokens to build leverage (§8).
3. **It earns native yield.** Yield-bearing collateral (aUSDC, wstETH) accrues its underlying return continuously; the ERC-4626 share price appreciates automatically.
4. **It absorbs risk.** It is the first-loss capital that stands behind redemption performance — an unresponsive maker, a depegged reserve — in exchange for a premium.

The jobs are configurable, not mandatory. Each vault opts into its risk roles independently: whether it stands behind forced redemptions (only the protocol-designated force-execute vault can be claimed against, §7) and whether it runs a lending market (a vault lends only once a borrow manager is bound to it, §8) are separate switches. That yields three vault profiles on one chassis — a pure *insurance vault* (force-executable, no lending book), a pure *lending vault* (earns the borrow premium but is never liquidated for redemptions), and the full *insurance + lending vault* that does both. LPs therefore choose a risk-yield point rather than accepting one bundle, and the protocol can match each collateral type to the job it suits — while every counted vault, whatever its profile, still contributes to the global solvency total of §6.

Multiple LPs deposit into a vault and receive **ERC-4626 shares** proportional to their contribution. There is one vault per collateral type (USDC, aUSDC, WETH, wstETH), and one vault manager per vault. Withdrawals are asynchronous by design: an LP requests an exit, waits out a short queue (72 hours at launch), and the exit clears only if releasing that collateral keeps global utilization inside the cap of §6 —

insurance cannot run away exactly when it is needed. Critically, **vaults are not fund-flow intermediaries**: minter stablecoins go to the market maker via the RFQ escrow, never through the vault. The vault holds LP collateral purely as trustless security. This is what lets the same dollar of collateral back issuance *and* earn yield *and* seed the lending book without being double-spent: the jobs are layered in risk, not in cash.

This is the second half of the thesis: **security and utility fund each other**. The lending book and the trading flow generate the premiums that pay LPs (§9); those premiums keep the insurance pool funded; and the funded pool lets the next CST launch with a lending market already underneath it. A CDP's collateral is a cost the minter pays. An MPV's collateral is a business.

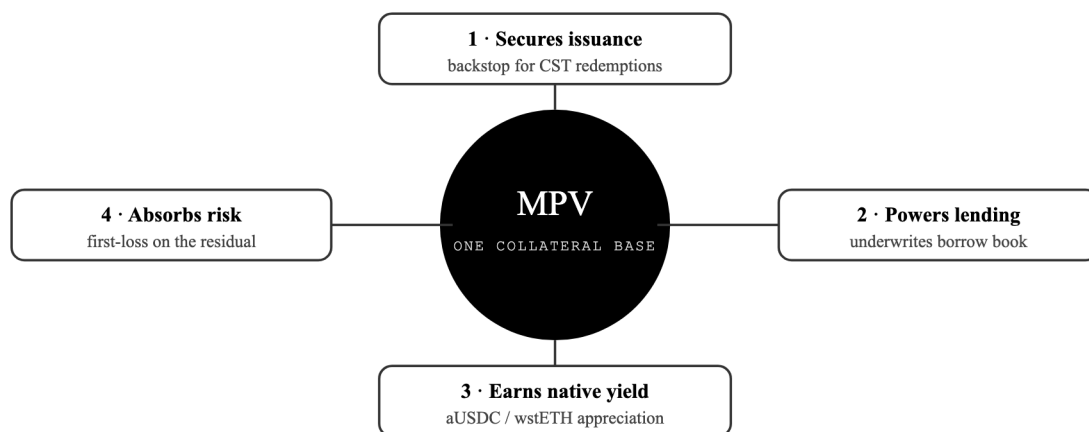


Figure 3. The Multi-Purpose Vault. A single ERC-4626 collateral base is layered across four non-conflicting roles. LP shares appreciate from native yield, the lending premium, and the flow share — projected at roughly twice the collateral's standalone benchmark (§9, Table A2).

POOLED INSURANCE, ISOLATED CUSTODY

Reserve Vaults are accounted *per asset*: each one backs exactly the eToken it is registered to, and nets against that asset's exposure alone. MPV solvency is accounted *globally* across all vaults (a mint anywhere draws on the protocol's total collateral and net exposure), but custody and yield are *isolated* per vault: each collateral type keeps its own LP shares, its own native yield, and its own lending book. A shortfall on any asset's residual is mutualized across the global pool, strengthening CST backing at the cost of shared LP risk (§12).

5. Issuance: RFQ & the PSM

A CST is issued through two first-class channels that feed one exposure ledger. The **RFQ marketplace** prices in cash: a market maker signs a firm quote offchain and settlement is atomic onchain. The **peg-stability module (PSM)** prices in kind: anyone converts a wrapper stock token into the eToken, and back, at a ratio derived from oracles — no quote, no fee, no permission.

5.1 RFQ: cash in, cash out

There is no onchain mint or redeem fee on the RFQ path: the market maker's margin is the bid/ask spread baked into the quote. A **quote** is an EIP-712 signed object binding the taker, asset, side, amount, price, a single-use quote id, and an expiry. The signed digest commits the chain id and market address to prevent cross-chain and cross-contract replay, and each digest is consumable once. A quote is accepted only if it recovers to a registered signer. Two settlement paths exist:

- **Market order:** the taker submits the signed quote and it settles atomically in one transaction. Nothing is persisted.
- **Limit order:** the taker rests an order onchain, escrowing the input; a maker later fills it (fully or partially) with a signed quote whose price satisfies the limit.

On a **mint**, the full payment-token amount is routed from the taker to the market maker's linked settlement address, and eTokens are minted to the taker:

$$q = \frac{a \cdot 10^{18}}{p} \quad (\text{decimal-adjusted}),$$

Eq. 5: eTokens minted for payment amount a at quoted price p

On a **redeem**, payment tokens flow from the market maker's linked address to the taker, the taker's eTokens burn, and global exposure shrinks. Decoupling the hot signing key from the funds custodian (the "linked address") bounds the damage a leaked key can do.

The backfill loop. An RFQ mint is carried by the crypto buffer only momentarily. The maker takes the buyer's stablecoins, buys the wrapper token in the primary or secondary market, and deposits it into the asset's Reserve Vault — a permissionless deposit that nets the new exposure to zero and frees the buffer for the next mint (§6). The mirror holds on redemptions: a maker that pays a redemption in cash may reclaim the matched wrapper the burn released, always clamped so the remaining reserve covers remaining exposure. The maker's working capital is therefore $\approx 1\times$ flow, not $3\times$ inventory: it recycles the same dollars through buy-wrapper-deposit cycles while the reserve, not the maker's balance sheet, holds the standing backing.

Price protection. Beyond the user's own limit, every market and limit settlement must price within a band of the asset's keeper-cached mark m :

$$|p - m| \leq \delta \cdot m, \quad \delta = \text{settleBandBps} \text{ (default 5\%)}. \quad \delta = \text{settleBandBps}$$

Eq. 6: settle band bounds a compromised signer

This caps the harm from a compromised signer key to δ of notional. Force-executed redemptions are exempt from the band because they price at the bare oracle limit (§7).

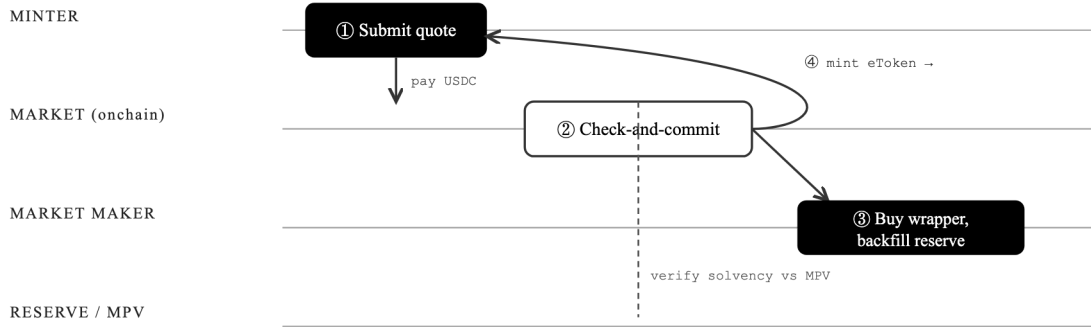


Figure 4. RFQ mint lifecycle. The market atomically checks new exposure against the global solvency caps (§6), routes payment to the maker's linked address, and mints the CST to the taker in a single transaction. The maker then buys the wrapper token and deposits it into the Reserve Vault; the netting of \$6 frees the crypto buffer, which carried the exposure only in the interim.

5.2 The PSM: wrapper in, eToken out

The PSM is the protocol's answer to MakerDAO's peg-stability module, generalized from a \$1 peg to a moving price. Any holder of a registered wrapper token can convert it into the corresponding eToken, and any eToken holder can convert back into the wrapper, directly against the Reserve Vault: permissionless, quote-less, and fee-free in both directions. The conversion ratio is never stored; it is derived at execution from two oracle reads:

$$\rho_w = \frac{P_w}{m_a}, \quad q_{\text{mint}} = w \cdot \rho_w, \quad w_{\text{redeem}} = \frac{q}{\rho_w},$$

Eq. 7: the PSM ratio — wrapper price over asset mark, derived, never stored

where P_w is the wrapper token's oracle price and m_a the asset's cached mark. For a total-return wrapper (one that reinvests dividends) ρ_w is exactly the issuer's accumulation multiplier, so the ratio self-adjusts through dividends and corporate actions with no parameter updates; both conversions round in the protocol's favor. Because the two legs are the same equity, a PSM conversion is value-neutral: the reserve and the liability change by the same USD amount, and the solvency ledger (§6) is untouched.

The PSM is also the arbitrage anchor the CDP lacked (§2.1). If the eToken trades rich against the wrapper on any venue, anyone mints through the PSM and sells; if it trades cheap, anyone buys and redeems in-kind. Resting limit orders can likewise be filled against the reserve by anyone — a permissionless market-making channel alongside the verified RFQ makers of §5.1 — in a single delivery-versus-payment step, with the treasury taking an admin-set share of the filler's spread over the mark — the only fee anywhere on the PSM, levied on the arbitrageur's edge, never on the backing.

Fail-closed guards. PSM mints require fresh marks on both legs. A per-operation move in ρ_w beyond a jump bound freezes the pair until an operator acknowledges the corporate action that caused it, so a mispriced oracle cannot be laundered through conversions. Each wrapper can be paused individually, or restricted to fills only. Redemptions are deliberately harder than mints: in-kind exit keeps working

outside market hours (both marks freeze at the same close) and even for a permanently halted asset, where the redeemer receives the halt price's worth of wrapper.

6. Pooled Solvency & Risk Accounting

All exposure, reserve, and collateral valuation lives in a single risk hub and is valued only at keeper-cached oracle marks, never trade prices. Reserves net against exposure per asset; three $O(1)$ caps protect solvency, checked and committed on the same path that mints.

Let u_a be the outstanding units of asset a , m_a its cached USD mark, R_a the marked USD value of its Reserve Vaults, and C_{USD} the total counted crypto collateral value across all MPVs. Because a reserve is delta-one in its own asset, it cancels exposure dollar-for-dollar — but only up to the exposure that actually exists, and never across assets. The protocol therefore maintains, as an $O(1)$ running total, the **net exposure**

$$E_{\text{net}} = \sum_a \max(0, u_a m_a - R_a),$$

Eq. 8: global net exposure — reserves net per asset, surplus never cross-subsidizes

and enforces the **global solvency invariant** on every risk-increasing mint:

$$E_{\text{net}} \leq U_{\text{max}} \cdot C_{\text{USD}}, \quad U_{\text{max}} = \text{globalMaxUtilizationBps}.$$

Eq. 9: the master solvency constraint

Eq. 8 is where the backing stack becomes one number. A fully reserved asset contributes zero: it rides the reserve and consumes no LP collateral, which is why the steady state of the backfill loop (§5.1) is cheap. A freshly minted, not-yet-backfilled slice, or a depegged wrapper, shows up instantly as positive net exposure and is overcollateralized by the crypto pool at $1/U_{\text{max}}$. The $\max(0, \cdot)$ is deliberate: a reserve surplus on one asset (dividend drift) cannot be counted as backing for another. Two further caps bound concentration and per-asset issuance:

$$u_a m_a \leq \text{cap}_a \quad (\text{per-asset ceiling}); \quad c_v \leq \kappa_v C_{\text{USD}} \quad (\text{collateral concentration}).$$

Eq. 10: per-asset ceiling and per-vault concentration cap

A per-asset ceiling of zero blocks minting that asset: a fail-safe default. The concentration cap κ_v counts at most a capped share of any single vault's collateral toward the global total, so no one collateral type can dominate the backing. All three are re-checked whenever a permissionless keeper refreshes a mark.

Mark freshness. Opening exposure (risk-increasing) requires the asset mark to have been pulled within a freshness bound τ (one hour at launch; PSM mints additionally require a fresh wrapper mark); new exposure cannot open against a stale price. Closing exposure and redemptions deliberately tolerate stale marks; they only reduce risk — which is also what keeps in-kind exit open outside market hours. Because the same code path both validates and commits the mint, the issuance check is correct by construction: there is no window in which a mint is approved but not booked.

WHY THIS BEATS THE CDP

In a CDP the solvency check is *per-position* and the collateral is posted by the minter. Here it is *global*, the standing backing is a productive reserve, and the buffer is posted by LPs. The minter pays full notional and receives full exposure (Eq. 5); the reserve nets that exposure to zero (Eq. 8); and the safety margin $U_{\max}$ on whatever is left is provided by yield-seeking LPs, not extracted from the person who wants the exposure. That is the capital-efficiency inversion of Table 1.

6.1 Sizing $U_{\max}$: the buffer is a claim-window VaR

$U_{\max}$ is not a style choice; it is sized by the guarantee's worst case. The guarantee must hold not at the last mark but at claim settlement: between the freshest accounting and the end of a forced redemption (the claim threshold T , plus proof staleness), collateral can fall while the asset mark rises. Let δ_c be the worst credible collateral drawdown over that window at the chosen confidence level, and δ_e the worst credible rise in the asset mark. Solvency at settlement, even at the cap $E = U_{\max}C$, requires $C(1 - \delta_c) \geq E(1 + \delta_e)$, i.e.

$$U_{\max} \leq \frac{1 - \delta_c}{1 + \delta_e}.$$

Eq. 11: the cap is a value-at-risk bound over the claim window

This is Eq. 2 inverted, applied only to the residual of Eq. 8, over a horizon that shrinks from open-ended to a bounded claim window: that is the quantitative payoff of layered backing. A CDP's ρ must survive the joint worst case per position, continuously; the CST buffer only has to survive the move between a default and its settlement. Worked conservatively: stablecoin collateral ($\delta_c \approx 0$) against a 10% overnight equity gap gives $U_{\max} \leq 0.91$; staked-ETH collateral at a 99th-percentile 48-hour drawdown ($\delta_c \approx 30\%$) against the same gap gives $U_{\max} \leq 0.64$. The deployed ladder (65% stablecoin, 55% BTC, 50% staked ETH) sits inside these bounds, and the slack is deliberate: it absorbs oracle staleness, liquidation slippage, and correlation between crypto drawdowns and redemption waves. The golden rule of §9 forbids spending that slack on growth.

7. The Redemption Guarantee

A CST holder has as many exits as the backing has layers. A market maker quotes cash; the Reserve Vault redeems in-kind, instantly and permissionlessly; and if both lapse, the crypto collateral is liquidated onchain, at the oracle price, to make the holder whole. In insurance terms, force execution is the claim process and the oracle-priced liquidation is the payout — and the claim is rarely needed precisely because the cheaper exits sit in front of it.

A redeemer holding eTokens is *captive* in a way a would-be minter is not: an unfilled mint simply leaves the user holding their stablecoins, but an unfilled redeem leaves the user holding a token they cannot exit. The waterfall answers that in order of cost. **Tier 1: a maker fills** the redeem in stablecoins at its quoted spread. **Tier 2: the PSM redeems in-kind** — whenever the reserve covers, any holder converts eTokens back into the wrapper token at the derived ratio (Eq. 7), with no quote, no waiting, and no dependence on any maker; it works outside market hours and even for a halted asset (§5.2). **Tiers 3–4: the claim** is the

recourse when the reserve is short and no maker fills. After a global claim threshold T (48 hours by default) elapses from order creation with no fill, the order owner may force the redemption against the designated MPV:

1. The named vault must equal the protocol-designated force-execute vault (else the call reverts); if no vault is designated, force execution is disabled, a fail-safe.
2. Fresh oracle proofs for the asset price P and collateral price P_c are supplied and verified (bounded staleness, not future-dated). The asset price must satisfy the order's limit floor.
3. The remaining q eTokens are valued at the user's own limit price (the bare oracle price, with no maker spread), converted to the vault's collateral, and released directly to the redeemer:

$$x = \frac{q \cdot P}{P_c} \quad (\text{collateral units released}),$$

Eq. 12: force-execution collateral release

the escrowed eTokens burn, and global exposure shrinks by qP . The guarantee follows directly from the netting of Eq. 8 and the invariant of Eq. 9: **so long as reserves plus counted collateral cover exposure — $R + C_{\text{USD}} \geq E_{\text{USD}}$, which Eq. 9 enforces with margin — every outstanding CST can be redeemed at full oracle value, in the wrapper through the PSM or in collateral by force**, whether or not any market maker participates. The maker is a convenience; the portfolio is the guarantee.

Claims are stabilising. A forced redemption of notional $x = qP$ removes the same dollar amount from exposure and from collateral, so global utilization strictly improves:

$$\frac{E_{\text{net}} - x}{C_{\text{USD}} - x} < \frac{E_{\text{net}}}{C_{\text{USD}}} \quad \text{whenever } E_{\text{net}} < C_{\text{USD}},$$

Eq. 13: a claim strictly deleverages the system

and the solvency cap guarantees the condition ($E_{\text{net}} \leq U_{\text{max}} C$ with $U_{\text{max}} < 1$). Every claim leaves the system better collateralised than it found it. Contrast the CDP fire sale of §2.2, where each liquidation makes the next more likely: the same stress that cascades a CDP book strictly deleverages a CST book.

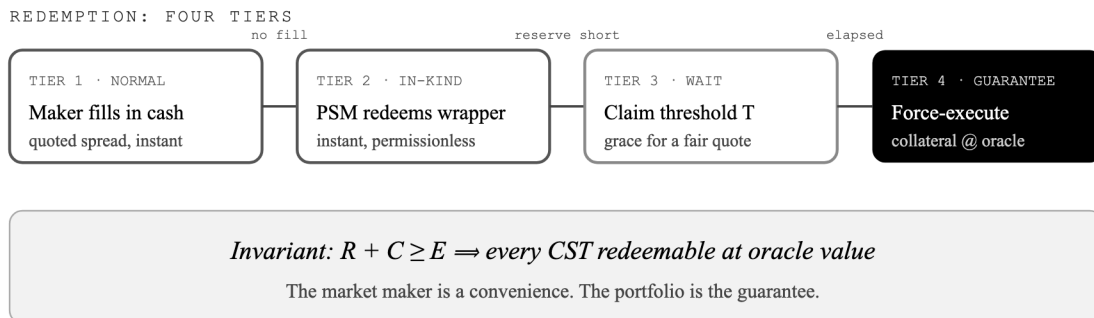


Figure 5. The four-tier redemption waterfall. Tier 1 relies on a market maker; Tier 2 is the PSM's instant in-kind exit against the Reserve Vault (Eq. 7); Tiers 3–4 are the trustless backstop that makes the holder whole from MPV collateral at the oracle price (Eq. 12), enforced by the solvency invariant (Eq. 9).

Emergency settlement. If an asset must be wound down permanently, an operator can halt it at a fixed settlement price, after which holders redeem at that price from a designated halt-redeem fund — and the PSM keeps honoring in-kind exits at the halt price for as long as reserve remains (§5.2). In-kind redemption, force execution, and halt redemption together ensure a holder can always exit, in every regime.

8. Lending & Leverage

Because a CST is a real ERC-20 backed by real collateral, it can be borrowed against. Each MPV runs a lending market where holders post eTokens as collateral and draw stablecoins, sourced from the vault's external credit line — Aave v3 [4] where it exists, the protocol's own minimal lending pool on chains where it does not — to loop into leveraged exposure. This is the same collateral base doing its second job.

8.1 Looping

A holder mints a CST, posts it as collateral, borrows stablecoins, mints more, and repeats. At a per-loop loan-to-value of θ (default 70%), the geometric series bounds terminal exposure per unit of equity:

$$L = \sum_{k=0}^{\infty} \theta^k = \frac{1}{1-\theta} \approx 3.3 \times \quad (\theta = 0.7).$$

Eq. 14: maximum loop leverage

8.2 The two-slope rate and the kink

Debt accrues through a single global interest index on a two-slope curve. Borrowers pay the greater of the live external base rate (Aave v3 [4]) or a floor, plus a utilization-based premium $\pi(u)$:

$$r(u) = \max(r_{\text{base}}, r_{\text{floor}}) + \pi(u), \quad \pi(u) = \pi_0 + \begin{cases} s_1 \frac{u}{u^*}, & u \leq u^* \\ s_1 + s_2 \frac{u - u^*}{1 - u^*}, & u > u^* \end{cases}$$

*Eq. 15: borrow rate; kink at target utilization u^**

with launch parameters $\pi_0 = 3\%$, kink $u^* = 80\%$, $s_1 = +1\%$, $s_2 = +72\%$. One dial sets both yield and supply. Below the kink, borrowing is cheap and the book fills, lifting LP income gently. Above the kink, the rate spikes, making leverage uneconomic so borrowers deleverage, an automatic brake that protects LPs and keeps their capital withdrawable. The external base rate passes through to the credit line at cost; the premium above it is swept to the vault manager and redistributed — a protocol treasury cut, the remainder to LPs as vault yield (raising the ERC-4626 share price). An index floor guarantees the book's debt never falls below the real external debt, so the credit line's carry can never land on LPs.

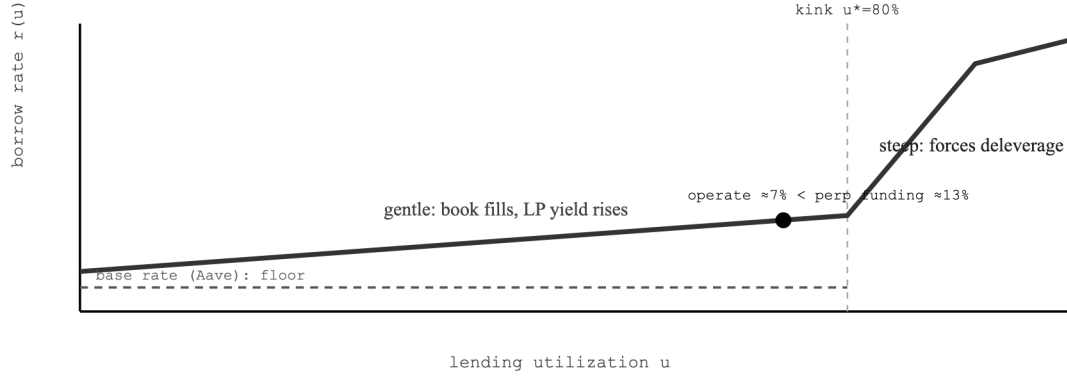


Figure 6. The two-slope rate of Eq. 15. The premium above the external base rate accrues to the vault manager and flows to LPs. The kink is placed where borrowing stays cheaper than perp funding, so the trade stays profitable for borrowers while paying LPs the most it safely can.

8.3 Health and self-healing liquidation

A position with n eTokens of collateral valued at P_e , against debt D and liquidation threshold ℓ (default 80%), has health factor

$$\text{HF} = \frac{n P_e \ell}{D}.$$

Eq. 16: borrower health factor

When $\text{HF} < 1$ a liquidator repays up to a close-factor cap and seizes collateral plus a bonus β (default 5%): $\text{seized units} = \text{repaid} \cdot (1 + \beta) / P_e$. Crucially, liquidating a CST borrower is **self-healing**: the burned/seized eTokens reduce protocol exposure E while the recovered stablecoins repay the vault's external debt; both health factors improve at once. This is the structural opposite of the reflexive CDP spiral of §2.2. In the rare case a position's collateral is exhausted before its debt, an operator-triggered absorption fronts the residual in cash, repays the credit line in full, and books the loss transparently against vault collateral — bad debt is settled the block it is recognized, never left to compound against LPs.

9. Economic Security & Incentives

A guarantee is only as strong as the incentive to provide the collateral behind it. The MPV is designed so that LP yield rises precisely when demand for the guarantee is highest: the system pulls in the collateral it needs, exactly when it needs it.

9.1 Where the demand comes from

The natural borrower is a delta-neutral fund harvesting perpetual-futures funding. Perp longs on hot names pay funding to shorts; a fund that holds an offsetting long does not care which way the stock moves and collects the funding. Own supplies that long cheaply and with leverage. With equity x , loop leverage L , borrow rate r , and perp funding f , the net annualized return on equity is

$$\text{Net} = x[Lf - (L - 1)r].$$

Eq. 17: funding-arbitrage net yield

For $x = \$100\text{k}$, $L = 3$, $f = 13\%$, $r = 7\%$: $\text{Net} = \$100\text{k}[3(0.13) - 2(0.07)] = \25k , a $\sim 25\%$ gross return that is indifferent to the direction of the underlying. This makes the fund a stable, repeat borrower, exactly what keeps the lending book full.

9.2 The self-balancing flywheel

The borrow rate is the bridge that pays LPs. Borrowers pay r ; the premium above the external base rate flows to LPs and the protocol. Because the premium rises with utilization (Eq. 15), and utilization rises with borrowing demand, **LP yield automatically increases when demand is hottest**, with no parameter changes. Higher yield attracts more collateral, which expands lending capacity, which admits more borrowers: a compounding flywheel. Capacity on the unreserved residual is linear in collateral ($E_{\max} = U_{\max} C$), and collateral follows yield, so the system scales without ever loosening a safety parameter. LP income stacks three layers:

$$r_{\text{LP}} = \underbrace{y}_{\text{native yield}} + \underbrace{\phi_{\pi} \frac{\pi(u) B}{C}}_{\text{lending premium}} + \underbrace{\phi_d \frac{d N}{C}}_{\text{flow share}}$$

Eq. 18: the LP yield stack (B borrowed, N custodied notional, d dividend/flow rate, ϕ revenue shares)

The third layer is the revenue the collateral's standing produces beyond lending: dividends accruing on eTokens custodied in the borrow book, and the protocol's share of PSM fill spreads, both routed to the vault through the same yield distributor as the lending premium. In the projected operating scenario the stack lands at roughly twice each collateral's standalone benchmark; the worked figures, and the assumptions behind them, are in Appendix Table A2 — projections, not promises.

THE GOLDEN RULE

Never raise the backing cap $U_{\max}$ to make room for more demand. The cap is the safety limit. When it binds, the fix is to attract more collateral (which raises the ceiling automatically), not to loosen the guarantee.

9.3 Why the market maker does not default

The guarantee makes holders whole *if* a maker defaults; it does not rely on makers choosing not to. The maker's own ledger does that. The standing backing lives in the Reserve Vault, not on the maker's balance sheet, so what a defaulting maker can extract is not the notional it has ever quoted but the unsettled flow of one claim window — and honoring a redemption in cash releases the matched reserve back to it (§5.1), so performing is itself compensated. What a default forfeits is the franchise: the discounted value of the spread business, plus everything recourse can reach:

$$\text{default pays only if } \Delta_{\text{window}} > \underbrace{\frac{sV}{r_d}}_{\text{franchise value}} + \Pi_{\text{recourse}}$$

Eq. 19: the maker's default condition (s spread, V annual volume, r_d discount rate)

Δ_{window} is small by construction when backing sits in the reserve, while sV/r_d grows with the very volume that would make defaulting tempting. Π_{recourse} is enforced offchain: quoting makers are protocol-onboarded entities with enumerated default triggers and recourse commitments (§12). And not all market making requires onboarding: anyone can act as a **permissionless market maker** by filling resting orders against the reserve through the PSM (§5.2) — those fills settle delivery-versus-payment in a single transaction, so there is nothing to default on. The protocol does not assume honest makers; it makes dishonesty a bad trade, insures against it anyway, and keeps a maker-free path open besides.

10. Who a CST Serves

A protocol clears when every participant gets something the alternatives do not offer. Each actor in this system has a distinct reason to show up, and each pays a visible price for it.

Table 3: what each participant gets, and what it costs them.

Participant	What they get	What they give up / risk
Holder / minter	Verifiable RWA exposure at 1:1 capital. No funding rate, no holding cost, no liquidation risk; self-custodial, with dividends passed through. A composable ERC-20 that works across DeFi.	Bid/ask spread on cash entry and exit; worst-case exit (reserve short, no maker) waits out the claim window (§7).
Borrower (delta-neutral fund)	The cheap leveraged long leg of the funding arbitrage (Eq. 17): up to $\approx 3.3\times$ via looping (Eq. 14) at a borrow rate kept below perp funding.	Pays the borrow premium that funds LP yield; carries liquidation risk on the loop (§8.3).
Liquidity provider	Underwriting income: native yield plus lending premium plus flow share, projected at roughly $2\times$ the collateral's standalone benchmark (Eq. 18, Table A2).	First-loss capital on the unreserved residual, mutualised across vaults (§12).
Market maker	The spread on every RFQ mint and redeem, with $\approx 1\times$ recycled working capital (§5.1) — the reserve, not its balance sheet, holds the inventory — on distribution rails it does not have to build.	Must honor redemptions and backfill reserves; a default forfeits the business and triggers the insurance liquidation.
Wrapper issuer (Ondo, Robinhood, ...)	A DeFi distribution channel: its tokens become productive reserve collateral inside a lending, leverage, and composability stack it does not have to build, held transparently onchain.	Its issuer risk is priced, capped, and diversified against — it is an ingredient, never the guarantee (§12).
DeFi at large	A new collateral class: equities, ETFs, and gold as permissionless, composable building blocks.	Inherits the trust assumptions of §12.

The common thread is access. Much of crypto-native capital sits beyond the reach of traditional brokerage infrastructure. Own gives that capital the exposure, the leverage, and the yield natively, in stablecoins,

onchain, with no custodian in the middle. The demand engine of §9 is not hypothetical users discovering stocks; it is existing capital that already wants this trade and has no other rail to run it on.

11. Oracle & Price Attestation

Every valuation in the protocol (marks for risk accounting, the PSM's derived ratio on both legs, force-execution payouts, the lending debt cap) is driven by an external oracle. Onchain spot prices (DEX reserves) are never used for valuation. Wrapper tokens held in Reserve Vaults are priced by the same machinery as the assets they back, which is what makes the netting of Eq. 8 and the ratio of Eq. 7 marks-consistent.

A dual-oracle architecture lets each asset select its price source:

- **In-house signed oracle.** An authorized signer submits an EIP-712 `PriceAttestation(asset, price, timestamp)`. Per-asset bounds reject stale prices and jumps beyond a deviation band, and timestamps must be strictly increasing. Emergency signer removal is instant.
- **Pyth Network [8].** Variable-exponent prices normalized to 18 decimals, gated by a confidence-interval bound and a max age, with up to four per-asset session feeds (regular, pre-market, post-market, overnight).

Both expose a cached read (for keeper marks) and an inline-proof read (for the force path). The oracle verifier is swappable behind the protocol registry, so the protocol can migrate to new price infrastructure (Chainlink, a ZK oracle) by updating a single address. Signed messages bind chain id and contract address to prevent replay; a compromised signer is bounded within the staleness and deviation bounds and the settle band of Eq. 6.

12. Trust Model & Limitations

The protocol assumes a small set of vetted privileged actors and treats end users as adversarial. The design goal is that no single actor can move value to itself, and that the onchain guarantee holds even if the market maker defaults.

Table 4: actors and trust levels.

Actor	Trust	Role & bound
Protocol admin	Trusted (7-day timelock)	Register contracts, assets, oracles; set global params.
Operator	Trusted (instant)	Emergency only: pause, halt, PSM ratio guard, remove signer; cannot move funds.
Wrapper issuer	Semi-trusted	Issues the reserve's stock tokens. A depeg or failure marks the reserve down; bounded by netting, per-wrapper pauses, and diversification.
Vault manager / maker	Semi-trusted	Quotes RFQ flow, backfills reserves, earns spread + premium. LP collateral backstops its default.
Quote / oracle signer	Semi / trusted	Sign quotes / prices offchain. Damage bounded by settle band, single-use quotes, deviation bounds.
Keeper	Untrusted	Permissionless: refresh marks, accrue interest, fulfil exits. Cannot move value to itself.
LP	Untrusted	Provides backstop collateral, earns yield. Capital at risk on VM default.
Minter / redeemer	Adversarial	Trades via escrow marketplace; redeemer has onchain force-execution recourse.

Accepted trade-offs. An auditor and an LP should weigh these deliberate choices:

- **Reserve issuer risk.** The reserve layer holds custodial wrapper tokens, so a slice of the backing inherits issuer risk — the very thing §1 criticizes. The difference is that here it is one priced ingredient, not the guarantee: a depeg marks the reserve down and reloads the residual onto the overcollateralized crypto layer automatically (Eq. 8), the ratio-jump guard freezes conversions on anomalous moves, each wrapper can be paused individually, and the roadmap diversifies reserves across issuers so no single one is ever load-bearing (§13).
- **Cross-vault loss mutualization.** A shortfall on any asset's residual is covered by the global collateral pool; all vaults' LPs share the loss. This strengthens CST backing (a bigger pool stands behind every token) at the cost of mutualizing LP risk across vaults.
- **Oracle signer trust.** The in-house oracle relies on a protocol-operated signer (single at launch, M-of-N on the roadmap). A compromised signer can mis-mark only within the staleness and deviation bounds.
- **Stablecoin 1:1 peg.** Lending math lifts stablecoins to USD at par; a depeg is invisible to debt valuation until marked.
- **Maker performance is attested, not proven.** Whether a maker will honor its next redemption and backfill its next mint is an offchain fact. This is why the reserve and the LP collateral — not the maker's promise — are the guarantee; real-world guards (protocol-onboarded makers, enumerated default triggers, recourse commitments) bound it in operation; permissionless PSM fills carry no such assumption at all, settling atomically (§5.2); and future phases add verifiable attestation of offchain holdings (§13).

Structural limitations. Beyond trust assumptions, four properties of the design itself that a holder or LP should price in:

- **Exit latency in the worst case.** A fully reserved asset exits instantly in-kind through the PSM. But if the reserve is short *and* no maker quotes, a redeemer waits out the claim threshold (48 hours by default) before force execution (§7). A CST guarantees exit *at oracle value*, not instant cash exit in every state; it is not a stablecoin-style at-par instrument.
- **Capacity is collateral-bounded.** Unreserved issuance can never exceed $U_{\max} C$ (Eq. 9). Growth is deliberately supply-constrained: when demand hits the cap, the protocol must attract more collateral or more reserve, never loosen the cap (§9). This trades growth speed for solvency.
- **The flywheel runs in reverse.** LP yield depends on borrowing demand. In a cold funding regime the premium compresses, collateral can leave, and residual capacity shrinks with it. The kink model slows the unwind (falling utilization makes borrowing cheaper, Eq. 15) but does not eliminate the cycle; the reserved slice of the backing is unaffected.
- **Price exposure, not ownership.** A CST confers the price of the asset, not the rights of a shareholder: no voting, no direct corporate actions. Synthetic exposure also faces jurisdictional restrictions that engineering cannot remove. The protocol layer stays permissionless and non-custodial; restrictions, where required, are applied at the distribution layer: by integrating apps, and by market makers, who choose whom they quote.

We state these plainly because a guarantee is only as credible as the failure modes it survives. The security posture still reduces to one line: any single layer of the backing — a wrapper issuer, a market maker, a collateral type — can fail, and the holder is made whole by the rest of the portfolio, enforced permissionlessly by the netting of Eq. 8 and the invariant of Eq. 9. That is the difference between trusting a custodian and holding a claim on a diversified, code-governed balance sheet.

13. The Decentralization Roadmap

DAI did not launch decentralized; it launched with a single collateral type and a founding team holding the oracle and emergency keys, and decentralized sleeve by sleeve as each new layer proved itself. A CST follows the same discipline: the solvency ledger of §6 prices every backing sleeve identically, so decentralization is a sequence of additions, never a redesign.

Phase 1 — launch: curated and small. The protocol launches deliberately concentrated so that every failure mode has a named owner: one reserve wrapper per asset where a quality issuer exists, a stablecoin MPV as the crypto layer, protocol-verified market makers, a protocol-operated oracle signer with Pyth as the second source, per-asset issuance caps in the low millions, and governance held by the team behind a timelock with instant-only-emergency operator powers. Small caps are the honest version of a new guarantee: the blast radius of anything unproven is bounded in dollars.

Phase 2 — diversify the portfolio. Each concentration is then widened where it is riskiest. Reserves spread across multiple wrapper issuers per asset (Ondo, Robinhood, Backed), so no single custodian backs a meaningful share of any token. The MPV set widens across collateral types (staked ETH, BTC), each behind its own concentration cap (Eq. 10). The oracle signer becomes an M-of-N set, and the verified-

maker roster opens to any firm that meets the recourse framework of §9.3 — alongside the permissionless PSM-fill channel, which is open to everyone from day one. Throughout, the golden rule of §9 holds: capacity grows by adding collateral and reserves, never by loosening $U_{\max}$.

Phase 3 — new sleeve types and open governance. The end state adds backing sleeves that today's architecture already prices but does not yet count: offchain hedges held by market makers, admitted as recognized backing once their holdings are verifiable (signed proof-of-reserves hardening into ZK proofs of offchain positions), and reserve assets beyond wrappers as tokenized market structure matures. Parameter authority — caps, the utilization ladder, asset onboarding — migrates from the team's timelock to token-holder governance, with the invariants of §6 as the constitution that governance cannot suspend, only tighten. At that point the eToken is what this paper set out to build: a decentralized stock token — and by composition, decentralized ETF and decentralized RWAs — backed by a self-balancing portfolio no single party can break.

14. Conclusion

We have described a system for real-world-asset exposure onchain that requires trust in no single custodian and imposes no overcollateralization tax on the holder. The CDP gave crypto trustless dollars but cannot give it trustless stocks: a stablecoin backs one volatile leg with collateral and pins the other at \$1, while an RWA forces you to back a volatile asset with another volatile one, which demands punitive, unscalable collateralization. Custodial tokenization tracks the price but concentrates the entire claim on one intermediary.

The Collateral-Secured Token resolves both the way DAI resolved the dollar: with a diversified collateral portfolio governed by code. Protocol-owned reserves of existing stock tokens back issued supply 1:1 and keep it delta-one; overcollateralized Multi-Purpose Vaults insure the residual while underwriting a lending market and earning native yield; and the PSM gives every holder two-way, fee-free convertibility between the token and its reserve. The redemption guarantee is not a promise from an issuer but a property of the code: as long as the portfolio covers exposure, any holder can redeem at oracle value — in-kind or in collateral — permissionlessly, whether or not any market maker, or any single custodian, participates.

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Appendix: Parameters & Notation

Table A1: key parameters and design defaults (all admin-configurable unless noted).

Symbol / param	Meaning	Default
$U_{\max}$ · globalMaxUtilizationBps	Global solvency cap: net exposure ÷ collateral	65% at launch
cap_a · assetCapUSD	Per-asset USD issuance ceiling	0 blocks mint; \$1M at launch
κ_v · collateralCapBps	Per-vault concentration cap	0 (uncapped)
δ · settleBandBps	Max settle deviation from mark	5%
τ · maxMarkAge	Mark freshness for opening exposure	1 h at launch
priceMaxAge	Inline oracle-proof freshness (force path, lending)	2 min
T · claimThreshold	Delay before force-execute allowed	48 h
ratioJumpBoundBps	PSM ratio-jump guard per operation	1.5%; 0 disables the PSM
psmFillSpreadShareBps	Treasury share of PSM fill spread	0
withdrawal wait	LP exit queue delay	72 h at launch
θ · borrowLtvBps	Per-position borrow LTV	70%
targetLtvBps	Vault-wide debt cap vs collateral	35%
ℓ · liquidationThresholdBps	Liquidation threshold	80%
β · liquidationBonusBps	Liquidator bonus	5%
u^* · kink	Target lending utilization	80%
π_0, s_1, s_2	Base premium / slope 1 / slope 2	3% / +1% / +72% at launch
timelock	Registry default-admin transfer delay	7 days

Table A2: projected LP yield stack at a healthy operating point. These are model projections, not a promise — real-world yields will differ with rates, utilization, and flow.

Income layer	aUSDC vault	wstETH vault	wBTC vault
Native base yield	3.1% (Aave)	2.6% (Lido)	~0% (idle BTC)
+ Lending premium (90% LP share)	+2.4%	+1.6%	+1.6%
+ Flow share (trading spread, dividends, PSM fills)	+2.0%	+2.0%	+2.0%
= Projected LP yield	~7.5%	~6.2%	~3.6%
vs. benchmark	~2.4× Aave	~2.4× stETH	vs ~0% idle

Assumptions. Insurance collateral at 50% of issued value (reserves carry the 1:1 backing); lending utilization 70% (aUSDC) / 50% (wstETH, wBTC) at the launch curve (Eq. 15), 90% of the premium to LPs, debt caps configured accordingly. Flow share: supply turns over ~5× a year at a ~0.4% round-trip spread (24/7 onchain trading, wide off-hours spreads), with about half routed to LPs. These are projections — realized yields will differ.

Standards compliance

ERC-20 + ERC-2612 (eToken); ERC-4626 (LP vault shares); ERC-7540 (async deposit/withdraw queues); EIP-712 (RFQ quotes and price attestations, domain "Own Protocol", v1, bound to chain id + contract).

Notation

P_t true asset price · $\tilde{P}_t$ synthetic market price · ρ CDP collateral ratio · e capital efficiency · E_{net} global net exposure · R_a reserve value of asset a · ρ_w PSM wrapper ratio · P_w wrapper price · C_{USD} global collateral · U_{max} utilization cap · m_a asset mark · q eTokens · p quote price · L loop leverage · θ borrow LTV · r borrow rate · $\pi(u)$ premium · u^* kink · HF health factor · β liq. bonus · f perp funding · δ settle band · δ_c, δ_e claim-window collateral drawdown / asset gap.